



Stony Brook Institute  
at Anhui University

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安徽大学纽约石溪学院

# ISE 333: User Interface Development (Design)

## Chapter 15: Information Search

## Chapter 16: Data Visualization

Spring 2026

Instructor: Hao Li

# Information Search

The home page of the U.S. Library of Congress Online Catalog (<https://catalog.loc.gov>) shows the simple search box prominently placed at the top of the page and provides alternative means of finding items of interest in the diverse collections. Advanced search interfaces are provided to accommodate experienced searchers.

The screenshot displays the Library of Congress Online Catalog homepage. At the top, the Library of Congress logo is on the left, and navigation links for 'ASK A LIBRARIAN', 'DIGITAL COLLECTIONS', and 'LIBRARY CATALOGS' are in the center. A search bar on the right contains the text 'Search Loc.gov' and a 'GO' button. Below the header, a dark banner reads 'Library of Congress > Library Catalogs'. The main content area is divided into a left sidebar and a main right section. The sidebar, titled 'LIBRARY OF CONGRESS ONLINE CATALOG', features a decorative image and a list of links: 'LC Online Catalog Home', 'About the Catalog', 'Frequently Asked Questions', 'Search/Browse Help', and 'Print/Save/Email Help'. Below this is a 'Search' section with links for 'Browse', 'Advanced Search', and 'Keyword Search', followed by a 'Your Account' section with links for 'Account Info' and 'Account Help'. The main right section is titled 'Library of Congress Online Catalog' and contains a 'LC Online Catalog Quick Search' box with a 'Search' button. To the right of the search box are links for 'Browse', 'Advanced Search', and 'Keyword Search'. Below the search box, a paragraph states: 'Contains 18 million catalog records for books, serials, manuscripts, maps, music, recordings, images, and electronic resources in the Library of Congress collections. Search LC Authorities at [authorities.loc.gov](https://authorities.loc.gov).' Further down, a section titled 'Additional Catalogs & Research Tools' explains that the LC Online Catalog is the main access point and provides links to specialized catalogs: 'Archival Finding Aids' (with a thumbnail image of a document), 'Copyright Office Catalog' (with a 'C' logo), 'LC Authorities' (with a thumbnail image of a classical statue), and 'E-Resources Online Catalog' (with a 'FENCO' logo). Each link is accompanied by a brief description of the resource.

LIBRARY OF CONGRESS

ASK A LIBRARIAN DIGITAL COLLECTIONS LIBRARY CATALOGS

Search Search Loc.gov GO

Library of Congress > Library Catalogs

LIBRARY OF CONGRESS ONLINE CATALOG

LC Online Catalog Home  
About the Catalog  
Frequently Asked Questions  
Search/Browse Help  
Print/Save/Email Help

Search  
Browse  
Advanced Search  
Keyword Search

Your Account  
Account Info  
Account Help

Library of Congress Online Catalog

LC Online Catalog Quick Search Search Browse Advanced Search Keyword Search

Contains 18 million catalog records for books, serials, manuscripts, maps, music, recordings, images, and electronic resources in the Library of Congress collections. Search LC Authorities at [authorities.loc.gov](https://authorities.loc.gov).

Additional Catalogs & Research Tools

The LC Online Catalog is the main access point to the Library's collections. Click on the links below to use specialized catalogs and tools that provide access to additional LC resources:

Archival Finding Aids  
Guides to unique Library of Congress archival collections

Copyright Office Catalog  
U.S. Copyright registrations and ownership documents, 1978-present

LC Authorities  
LC authority headings for subjects, names, titles, and name/title combinations

E-Resources Online Catalog  
Subscription and free databases, ejournals, and ebooks accessible at the Library

# Information Search

The screenshot displays the 'Advanced Search' page of the Library of Congress Online Catalog. At the top, the Library of Congress logo is on the left, and navigation links for 'ASK A LIBRARIAN', 'DIGITAL COLLECTIONS', and 'LIBRARY CATALOGS' are in the center. A search bar on the right contains the text 'Search Loc.gov' and a 'GO' button. Below the header, a breadcrumb trail reads 'Library of Congress > LC Online Catalog > Advanced Search'. On the right side of the page, there are links for 'Print', 'Subscribe', and 'Share/Save'. A horizontal menu bar includes 'LC Online Catalog', 'Browse', 'Advanced Search' (which is highlighted), and 'Keyword Search'. Below this, a secondary menu bar contains 'Search History', 'Account Info', 'Help', and 'LC Authorities'. The main section is titled 'Advanced Search' and contains several search criteria sections: 1. 'Search' section with three rows of input fields. The first row has 'user interface' in the text field, 'as a phrase' in a dropdown, and 'within' in another dropdown. The second row has a 'Keyword Anywhere (GKEY)' dropdown. Below this are radio buttons for 'AND', 'OR', and 'NOT'. The third row has 'video' in the text field, 'all of these' in a dropdown, and 'within' in another dropdown, followed by another 'Keyword Anywhere (GKEY)' dropdown. The fourth row has 'Hollywood' in the text field, 'all of these' in a dropdown, and 'within' in another dropdown, followed by another 'Keyword Anywhere (GKEY)' dropdown. 2. 'Remove Limits' link. 3. 'Year Published/Created' section with radio buttons for 'Year' and 'All Years', and 'From' and 'To' date fields. 4. 'Location in the Library' section with a dropdown menu showing 'All Locations', 'General Collections', 'Reference Collections, ALL', 'African Reference Collection', and 'African/Middle Eastern'. 5. 'Place of Publication' section with a dropdown menu showing 'All Places', 'Afghanistan', 'Alabama', 'Alaska', and 'Albania'. 6. 'Type of Material' section with a dropdown menu showing 'All Types', 'All Text (Books, Periodicals, etc.)', 'Archival Manuscript/Mixed Formats', 'Book', and 'Film or Video' (which is highlighted). 7. 'Language' section with a dropdown menu showing 'All Languages', 'English', 'Abkhaz', 'Achinese', and 'Acoli'. At the bottom, there is a 'Records per page' dropdown set to '25', a 'Clear' button, and a 'Search' button. Below these is an 'Advanced Search Tips' section with the following text: '1. Enter your search term(s) in the Search box(es):' and a sub-point '• Capitalization does not matter.'

Library of Congress

ASK A LIBRARIAN DIGITAL COLLECTIONS LIBRARY CATALOGS

Search Search Loc.gov GO

Library of Congress > LC Online Catalog > Advanced Search

Print Subscribe Share/Save

LC Online Catalog Browse Advanced Search Keyword Search

Search History Account Info Help LC Authorities

### Advanced Search

Search

user interface as a phrase within

Keyword Anywhere (GKEY)

AND OR NOT

video all of these within

Keyword Anywhere (GKEY)

AND OR NOT

Hollywood all of these within

Keyword Anywhere (GKEY)

[Remove Limits](#)

Year Published/Created

Year All Years From To

Location in the Library

All Locations  
General Collections  
Reference Collections, ALL  
African Reference Collection  
African/Middle Eastern

Place of Publication

All Places  
Afghanistan  
Alabama  
Alaska  
Albania

Type of Material

All Types  
All Text (Books, Periodicals, etc.)  
Archival Manuscript/Mixed Formats  
Book  
Film or Video

Language

All Languages  
English  
Abkhaz  
Achinese  
Acoli

Records per page: 25 Clear Search

Advanced Search Tips

1. Enter your search term(s) in the Search box(es):

- Capitalization does not matter.

The advanced search interface of the U.S. Library of Congress Online Catalog (<https://catalog.loc.gov>). The entire page is now dedicated to search controls and tips. Using check boxes, text fields, and menus, users can compose Boolean queries, restrict the search scope to a subset of the collections, and apply filters based on metadata. Regular users sign up for an account to save results and keep a search history to facilitate re-finding.



# Information Search

- Five-Stage Search Framework
  - Formulation
    - Expressing the search
  - Initiation of action
    - Launching the search
  - Review of results
    - Reading messages and outcomes
  - Refinement
  - Use
    - Compiling or disseminating insight

## 1. Formulation

- Use simple and advanced search.
- Limit the search using structured fields such as year, media, or location.
- Recognize *phrases* to allow entry of names, such as "George Washington."
- Permit *variants* to allow relaxation of search constraints (e.g., phonetic variations).
- Control the size of the initial result set.
- Use scoping of source carefully.
- Provide suggestions, hints, and common sources.

## 2. Initiation of action

- Explicit actions are initiated by buttons with consistent labels (such as "Search").
- Implicit actions are initiated by changes to a parameter and update results immediately.
- Guide users to successful or past queries with auto-complete.

## 3. Review of results

- Keep search terms and constraints visible.
- Provide an overview of the results (e.g., total number).
- Categorize results using metadata (by attribute value, topics, etc.).
- Provide descriptive previews of each result item.
- Highlight search terms in results.
- Allow examination of selected items.
- Provide visualizations when appropriate (e.g., maps or timelines).
- Allow adjustment of the size of the result set and which fields are displayed.
- Allow change of sequencing (alphabetical, chronological, relevance ranked, etc.).

## 4. Refinement

- Guide users in progressive refinement with meaningful messages.
- Make changing of search parameters convenient.
- Provide related searches.
- Provide suggestions for error correction (without forcing correction).

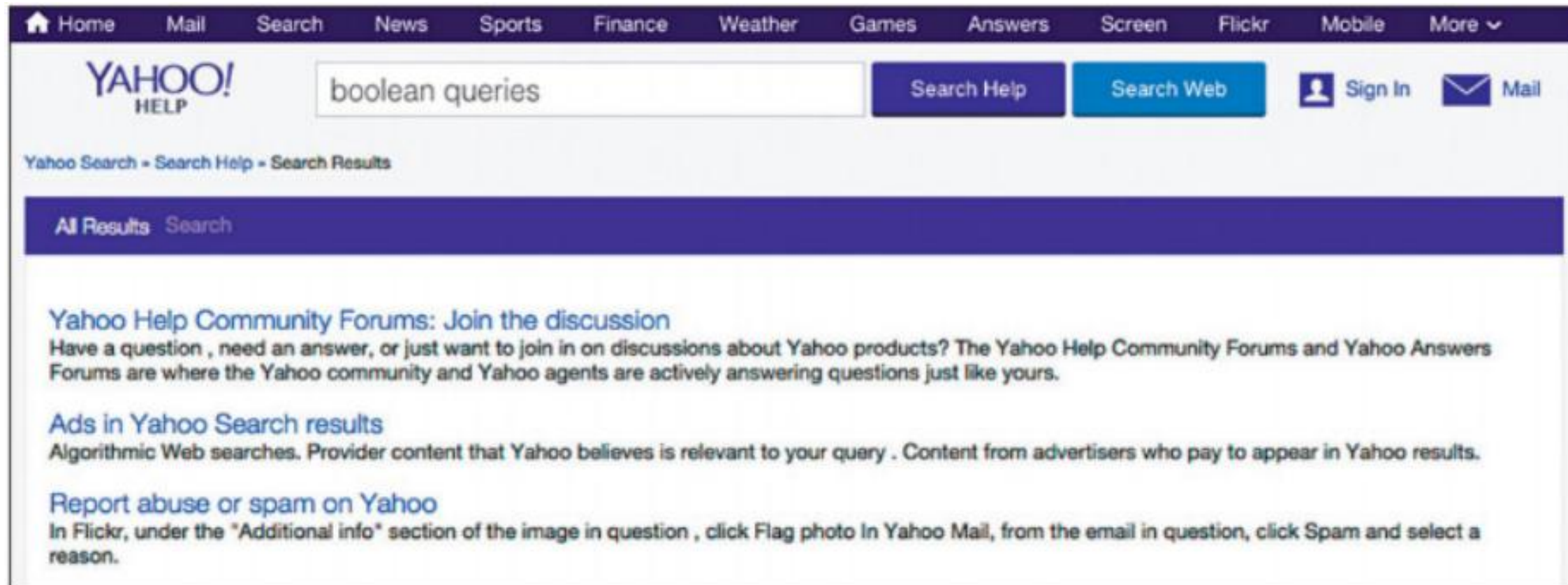
## 5. Use

- Embed actions in results when possible.
- Allow queries, settings, and results to be saved, annotated, and sent to other applications.
- Explore collecting explicit feedback (ratings, reviews, like, etc.).

# Information Search

- Formulation

The Yahoo! help search box has two buttons of different colors to search two different sources of information: purple for searching the help information and blue for searching the web. Pressing the purple button “scopes” the results to the help information only and shows results below a purple banner. Searching the web jumps to a different page (the normal search) that reuses the blue button color, helping users keep track of which source of information they are searching.

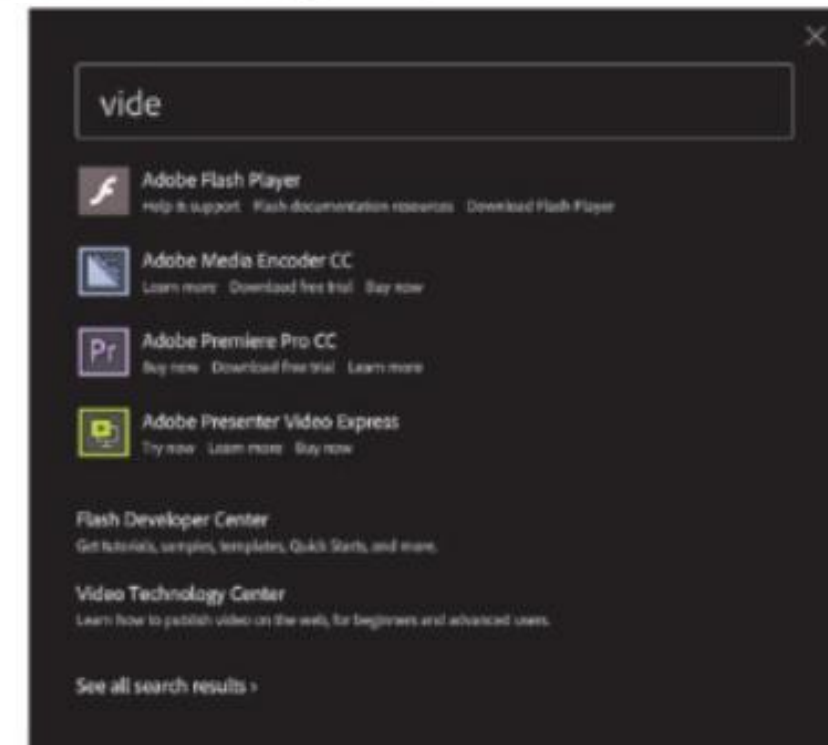
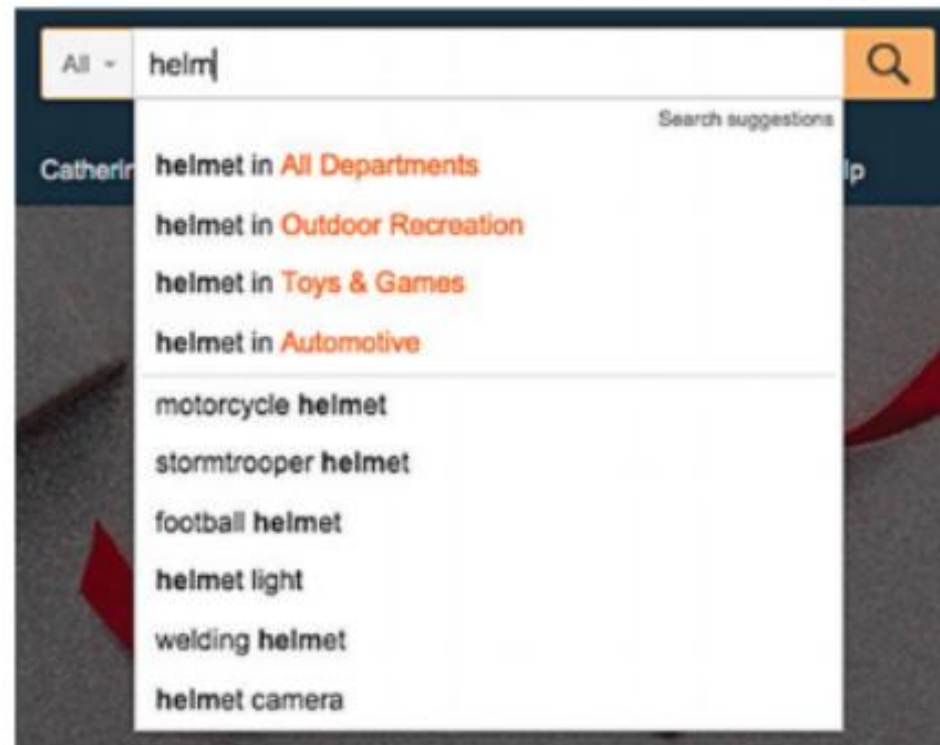


# Information Search

- Formulation

Auto-complete suggestions can speed data entry and guide users toward successful queries.

- (a) In a mobile phone address book, typing one character filters the list to all names that contain that character, and the list is updated continuously as users type.
- (b) Typing "helm" in Amazon's search box shows suggestions for "helmet light" or "welding helmet" but also suggestions to narrow the scope of the search to relevant departments.
- (c) On the Adobe website, suggestions include products (e.g., typing the beginning of the word "video" suggests several video-editing tools).



# Information Search

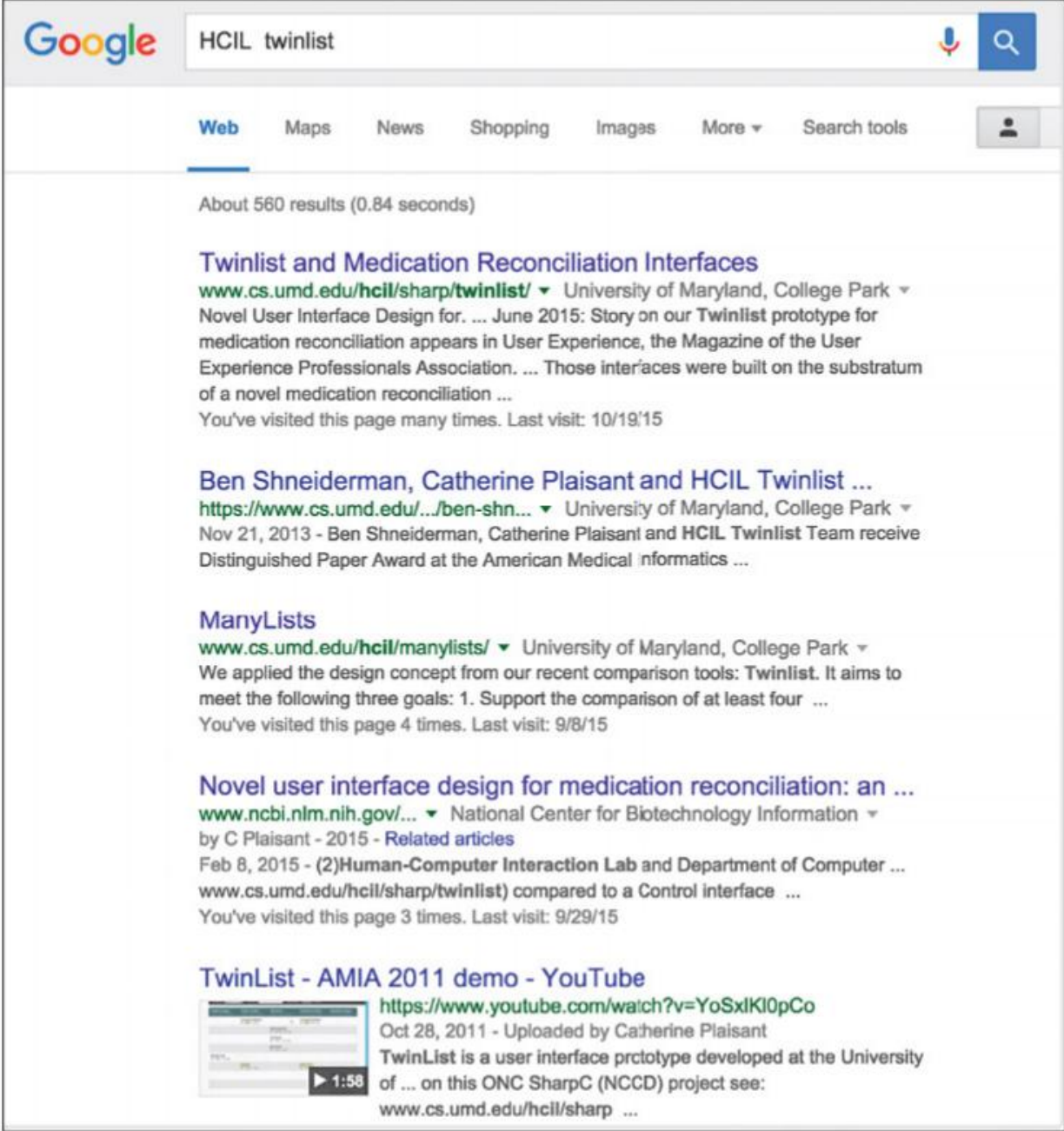
- Initiation of action
  - Explicit initiation
    - Click on a search button (or press the Enter key on a keyboard) to initiate the search
  - Implicit initiation
    - Each change to any component of the formulation stage immediately produces a new set of search results
    - Dynamic queries enable users' adjustment of query widgets to produce continuous updates



# Information Search

- Review of results

A Google Search result list. A summary is provided at the top (the total number of results). Each result includes preview information (or a snippet). Search terms are highlighted, including "Human-Computer Interaction Lab," which is the expanded variant of the search term "HCIL." The name of the top-level organization was added (here "National Center for Biotechnology Information") to help users judge the trustiness of the information.



The screenshot shows a Google search results page for the query "HCIL twinlist". The search bar at the top contains the text "HCIL twinlist". Below the search bar, there are tabs for "Web", "Maps", "News", "Shopping", "Images", "More", and "Search tools". The "Web" tab is selected. The results section shows "About 560 results (0.84 seconds)". The first result is titled "Twinlist and Medication Reconciliation Interfaces" and is from the University of Maryland, College Park. The second result is titled "Ben Shneiderman, Catherine Plaisant and HCIL Twinlist ..." and is also from the University of Maryland, College Park. The third result is titled "ManyLists" and is from the University of Maryland, College Park. The fourth result is titled "Novel user interface design for medication reconciliation: an ..." and is from the National Center for Biotechnology Information. The fifth result is titled "TwinList - AMIA 2011 demo - YouTube" and is a video from the University of Maryland, College Park.

Google HCIL twinlist

Web Maps News Shopping Images More Search tools

About 560 results (0.84 seconds)

**Twinlist and Medication Reconciliation Interfaces**  
[www.cs.umd.edu/hcil/sharp/twinlist/](http://www.cs.umd.edu/hcil/sharp/twinlist/) University of Maryland, College Park  
Novel User Interface Design for. ... June 2015: Story on our Twinlist prototype for medication reconciliation appears in User Experience, the Magazine of the User Experience Professionals Association. ... Those interfaces were built on the substratum of a novel medication reconciliation ...  
You've visited this page many times. Last visit: 10/19/15

**Ben Shneiderman, Catherine Plaisant and HCIL Twinlist ...**  
<https://www.cs.umd.edu/.../ben-shn...> University of Maryland, College Park  
Nov 21, 2013 - Ben Shneiderman, Catherine Plaisant and HCIL Twinlist Team receive Distinguished Paper Award at the American Medical Informatics ...

**ManyLists**  
[www.cs.umd.edu/hcil/manylists/](http://www.cs.umd.edu/hcil/manylists/) University of Maryland, College Park  
We applied the design concept from our recent comparison tools: Twinlist. It aims to meet the following three goals: 1. Support the comparison of at least four ...  
You've visited this page 4 times. Last visit: 9/8/15

**Novel user interface design for medication reconciliation: an ...**  
[www.ncbi.nlm.nih.gov/...](http://www.ncbi.nlm.nih.gov/...) National Center for Biotechnology Information  
by C Plaisant - 2015 - [Related articles](#)  
Feb 8, 2015 - (2)Human-Computer Interaction Lab and Department of Computer ...  
[www.cs.umd.edu/hcil/sharp/twinlist/](http://www.cs.umd.edu/hcil/sharp/twinlist/) compared to a Control interface ...  
You've visited this page 3 times. Last visit: 9/29/15

**TwinList - AMIA 2011 demo - YouTube**  
<https://www.youtube.com/watch?v=YoSxIKl0pCo>  
Oct 28, 2011 - Uploaded by Catherine Plaisant  
TwinList is a user interface prctotype developed at the University of ... on this ONC SharpC (NCCD) project see:  
[www.cs.umd.edu/hcil/sharp](http://www.cs.umd.edu/hcil/sharp) ...



# Information Search

- Review of results

Searching for Annapolis on the real estate website Zillow returns a list of houses and dots displayed on a map. The two windows are coordinated; when the cursor hovers over a house in the result list, the location of the house is indicated on the map. A click on the house would bring all the details displayed in an overlapping window.

The screenshot displays the Zillow website interface for a search in Annapolis, MD. The top navigation bar includes links for Buy, Rent, Sell, Mortgage, Agent Finder, Advice, Home design, and More. The search bar shows 'Annapolis' with filters for Listing Type, Price Range (\$100K - \$325K), Beds (0+), Home Type, and More. The map on the left shows numerous red and blue dots representing listings. A tooltip for a specific listing is visible: \$275K, 3 bd, 2 ba, 2,400 sqft. The right sidebar, titled 'Annapolis MD Real Estate', shows 489 results, 1 unmapped, and lists featured properties:

- 856 Barbud Dr, Crownsville**  
• HOUSE FOR SALE  
\$325,589  
4 bds · 2 ba · 1,976 sqft  
Special Offer: \$6,500
- 234 Main St, Annapolis, MD**  
• HOUSE FOR SALE  
\$275,000  
3 bds · 2 ba · 1,364 sqft  
Open: Sun. 3-5pm
- 2161 Greenway Crossing, Annapol...**  
• FORECLOSURE  
\$299,900  
5 bds · 1 ba · 1,456 sqft  
Less than 1 day on Zillow
- 9458 Doug's Court APT 302, Annapol...**  
• CONDO FOR SALE  
\$249,000  
3 bds · 2 ba · 1,328 sqft  
1 day on Zillow
- 5611 Mossy Parkway, Annapolis, MD**

# Information Search

- Review of results

A search for “human computer interaction” powered by Summon™ for a university library catalog returns a very large number of results. On the left, users can see the number of results for categories organized by content type, subject terms, or publication date. The box provides an overview of the results, reveals how the search was done (e.g., here the default search does not return dissertations), and facilitates further refinement of the search. The menu at the upper right allows users to sort results by relevance or by date. Help is available with a “Chat now” button, which allows users to chat with a librarian (<http://www.lib.ncsu.edu>).

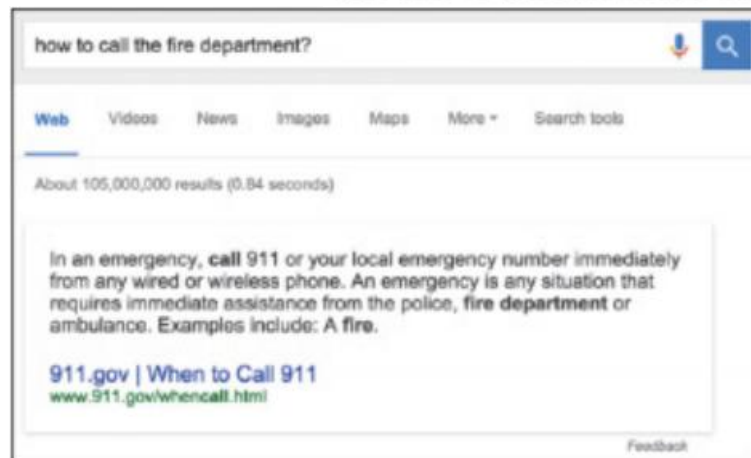
The screenshot shows the NCSU Libraries Summon search results page for the query "human computer interaction". The page is organized into several sections:

- Header:** NCSU LIBRARIES Summon logo, search bar with the query "human computer interaction", and navigation links (Help, About, chat now, English).
- Search Results:** A message stating "Your search for human computer interaction returned 905,538 results".
- Refine your search:** A sidebar on the left with filters for content type, subject terms, and publication date. The content type filter is expanded, showing options like "Journal Article (709,418)", "Book / eBook (596,491)", "Dissertation (318,738)", "Conference Proceeding (187,363)", "Book Chapter (144,577)", "Newspaper Article (92,884)", "Magazine Article (33,686)", "Trade Publication Article (30,821)", and "Journal / eJournal (98)".
- Recommendation:** A box titled "Recommendation: We found one or more specialized collections that might help you." with a link to "ACM Digital Library - Collection of citations and full text from ACM journal and newsletter articles and conference proceedings".
- ACM transactions on computer-human interaction:** A section featuring a book icon and text: "by Association for Computing Machinery", "ACM transactions on computer-human interaction, ISSN 1557-7325, 1998", "Human-computer interaction", and links for "eJournal: Full Text Online", "Journal: Check Availability", and "eJournal: Full Text Online".
- Human-computer interaction:** A section featuring a book icon and text: "Human-computer interaction, ISSN 1532-7051", "System design, Computers, Human-machine systems, Psychological aspects", and links for "eJournal: Full Text Online" and "Journal: Check Availability".
- Advances in human-computer interaction:** A section featuring a book icon and text: "Advances in human-computer interaction, ISSN 1687-5893", "Human-computer interaction", and links for "eJournal: Full Text Online" and "eJournal: Full Text Online".
- International journal of human-computer interaction:** A section featuring a book icon and text: "International journal of human-computer interaction, ISSN 1532-7590, 1989", "Human-computer interaction, Datorarbete, Människa-maskin-system, periodika, Människa-dator-interaktion", and links for "Journal: Check Availability", "eJournal: Full Text Online", and "eJournal: Full Text Online".
- Publication Date:** A bar chart showing the distribution of results by publication date, with a label "Any".
- Footer:** "2015 NCSU Libraries: Summon | Powered by Summon™", "Personalized Search", and "Saved Items (0)".

# Information Search

- Refinement
  - Search UIs can provide meaningful messages to explain search outcomes and to support progressive refinement
- Use

When possible (and important), provide information or simple actions without requiring users to leave the search results page. On the left, Google Search users get the answer to their safety-critical question at the top of the result list. On the right, Peapod shoppers looking for groceries can specify quantity and buy directly from the list of results after a search on “grapes.”



90 Results for: "grape"

Sorted by: Best Match

View as: Grid List

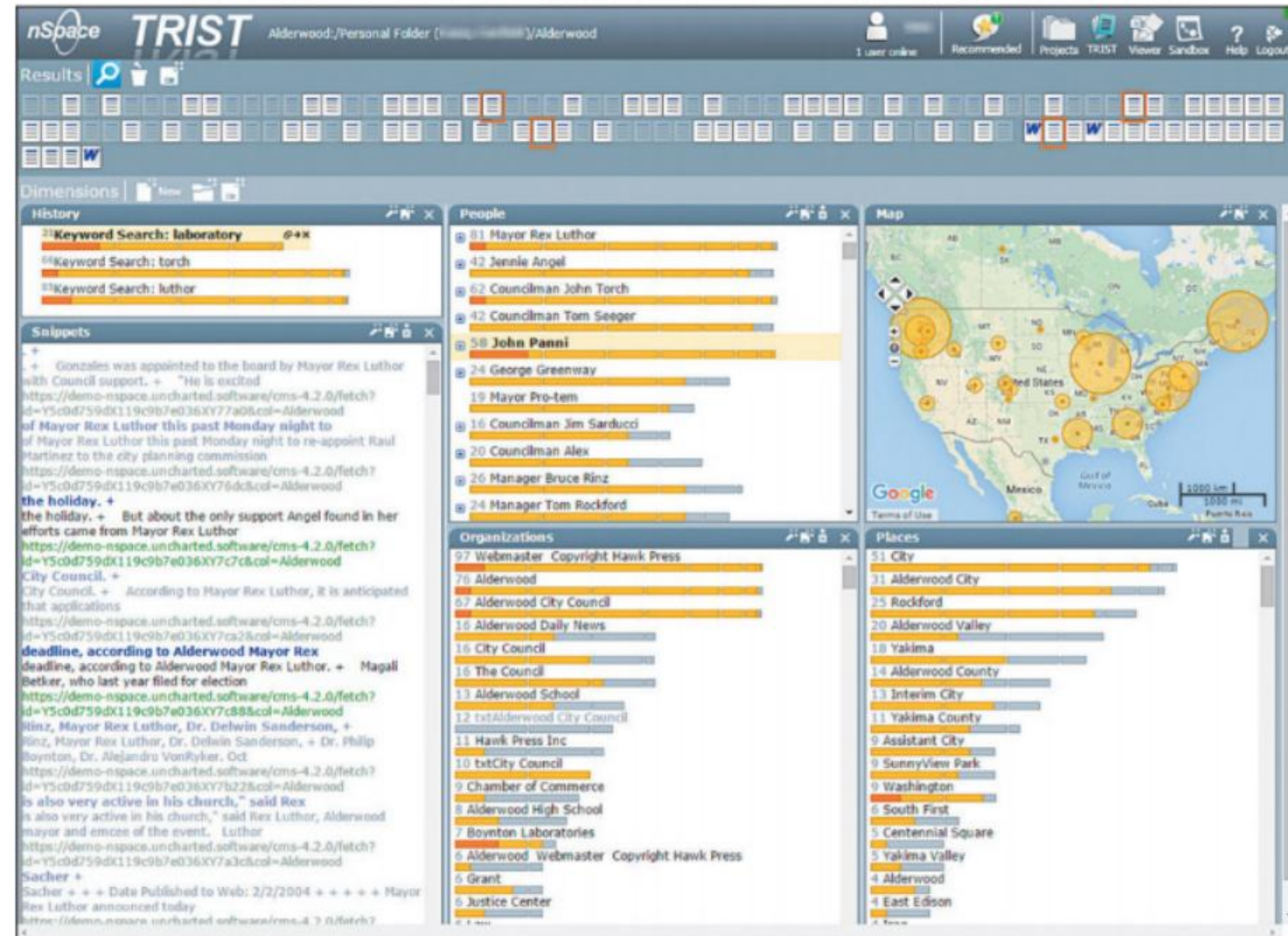
| Qty                                                                                                                  | Description                                                                                                         | Size     | Unit Price    | Price  |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------|---------------|--------|
|  <input type="text" value="1"/> |  Grapes Red Seedless Organic   | APX 2 LB | (\$4.99 / LB) | \$9.98 |
|  <input type="text" value="2"/> |  Grapes Green Seedless Organic | APX 2 LB | (\$4.99 / LB) | \$9.98 |



# Information Search

- Use

nSpace TRIST from Uncharted Software is used by analysts to produce evidence-based reports. For this (fictitious) criminal investigation, a user is reviewing a collection of documents (shown as icons at the top). The search history on the left shows three searches. Names, places, and organizations have been automatically extracted. Here the search term "Laboratory" and the person "John Panni" are selected. Snippets are shown on the bottom left. Analysts use TRIST to define dimensions of interest and to quickly identify documents of interest for use in a Sandbox (Fig. 15.10) (<http://uncharted.software/nspace>; Chien et al., 2008).

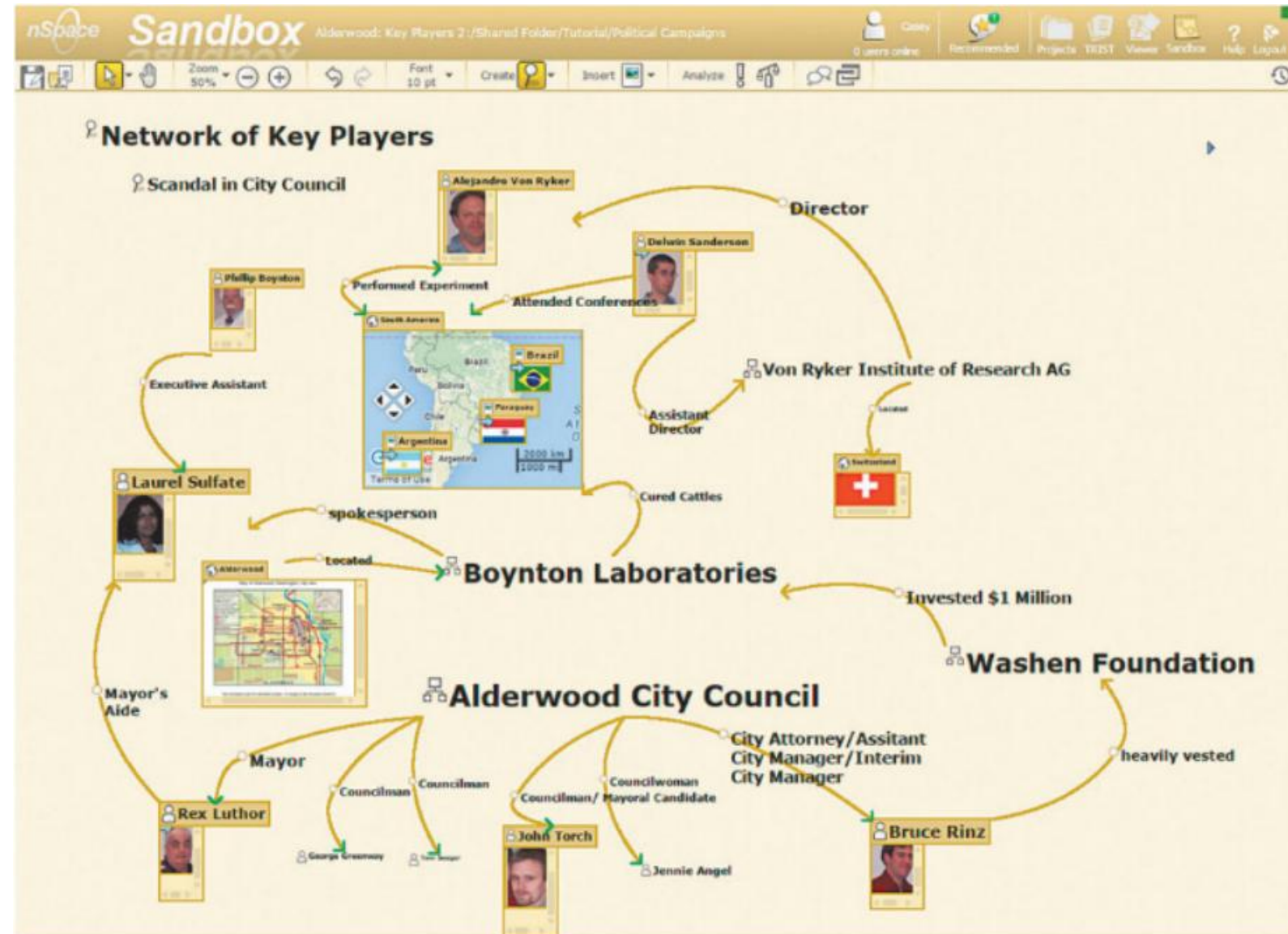




# Information Search

- Use

nSpace Sandbox<sup>®</sup> from Uncharted Software<sup>™</sup> allows multiple analysts to organize and present the evidence gathered from research. A variety of tools such as node and link diagramming, automatic source attribution, recursive evidence marshaling, and timeline construction provide support for analysis and reporting.



# Information Search

- Miscellaneous Modalities
  - Command language based search
  - Natural language based search
- Miscellaneous Data
  - Image search
  - Video search
  - Audio search
  - Geographic information search

# Data Visualization

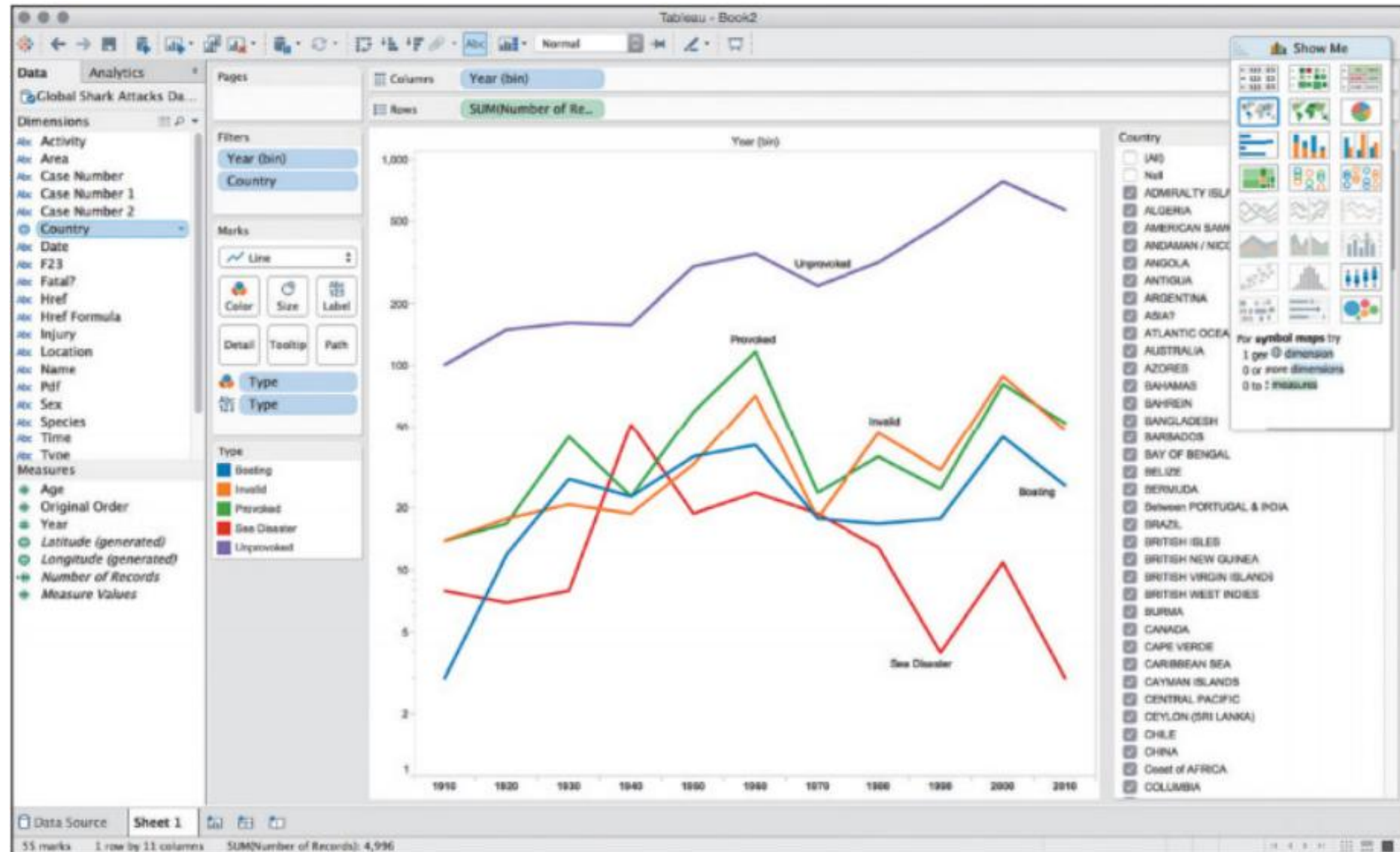
- Tasks in Data Visualization

| <b><i>Task Categories</i></b>      | <b><i>Task Types</i></b>                                                                                                                                                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Data and view specification</i> | <b>Visualize</b> data by choosing visual encodings<br><b>Filter</b> out data to focus on relevant items<br><b>Sort</b> items to expose patterns<br><b>Derive</b> values of models from source data                                            |
| <i>View manipulation</i>           | <b>Select</b> items to highlight, filter, or manipulate<br><b>Navigate</b> to examine high-level patterns and low-level detail<br><b>Coordinate</b> views for linked exploration<br><b>Organize</b> multiple windows and workspaces           |
| <i>Process and provenance</i>      | <b>Record</b> analysis histories for revisitation, review, and sharing<br><b>Annotate</b> patterns to document findings<br><b>Share</b> views and annotations to enable collaboration<br><b>Guide</b> users through analysis tasks or stories |



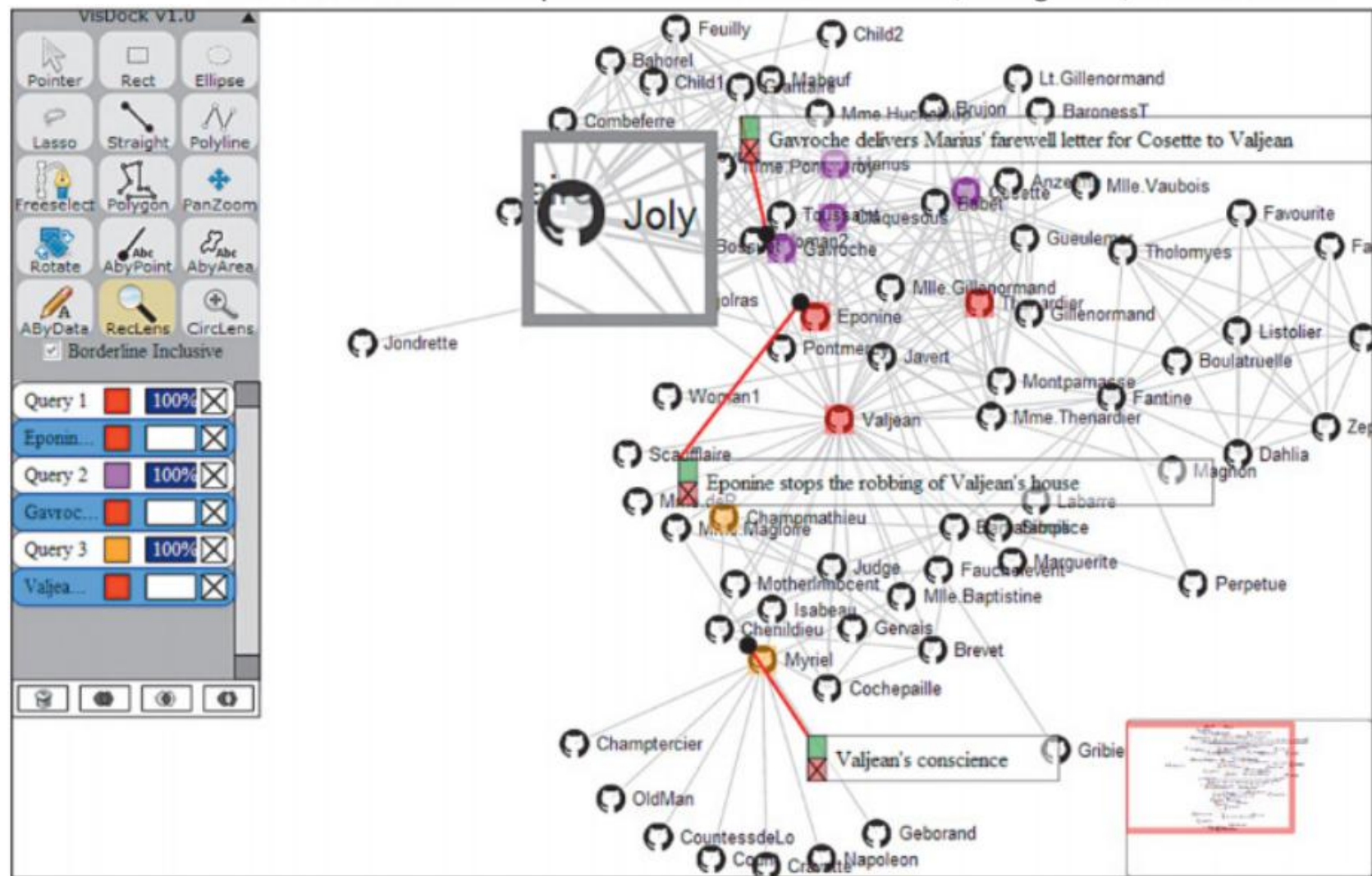
# Data Visualization

Visualization palette (upper right) in the Tableau Desktop application for a dataset of shark attacks. The “show me” feature in the tool (Mackinlay et al., 2007) will automatically highlight the suitable charts that can be used for the selected data.



# Data Visualization

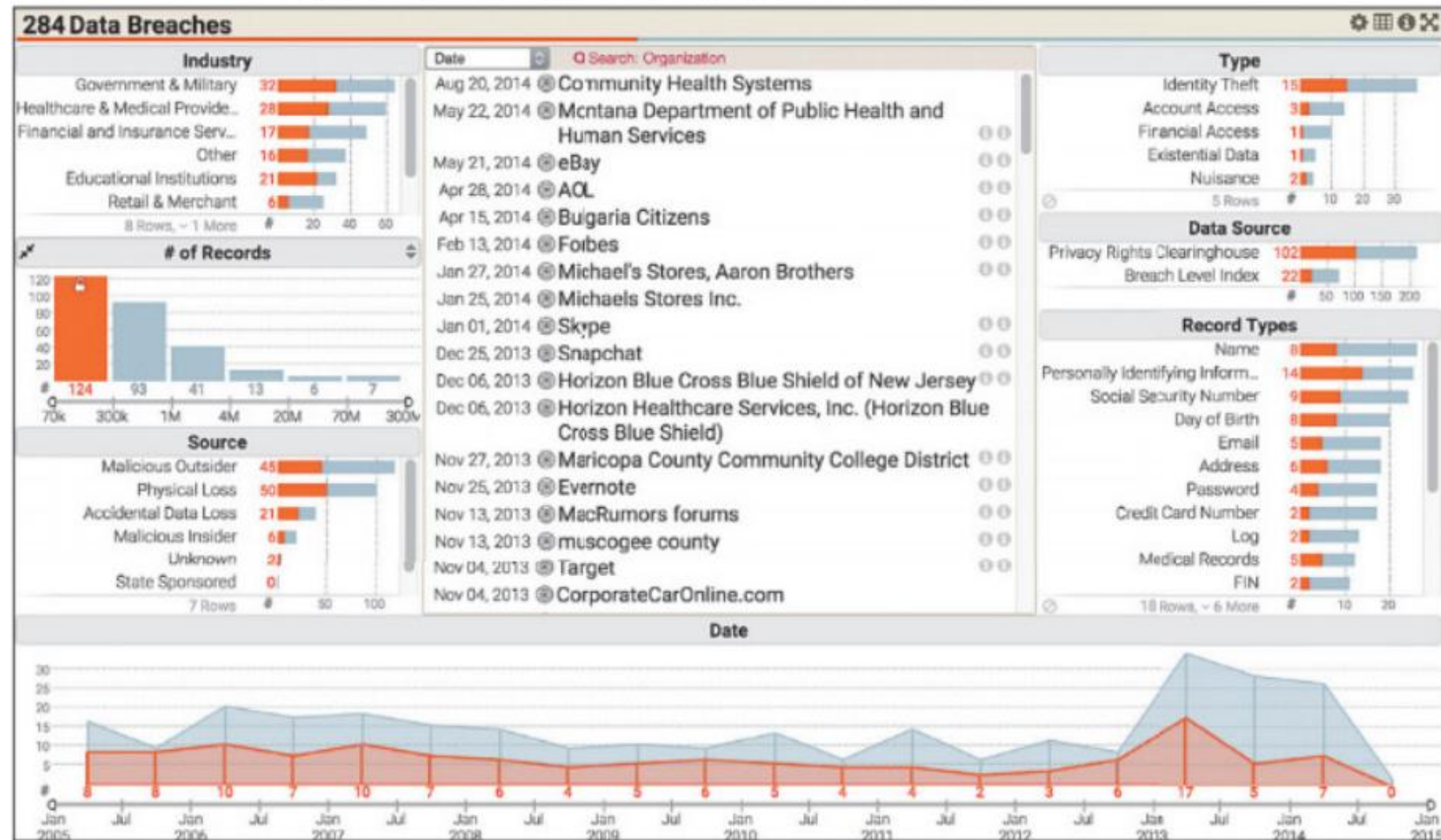
Selection tools and data-aware annotations in an interactive node-link diagram representation of the social network for all of the characters in Victor Hugo's *Les Misérables*. Characters are linked together if they appear in the same chapter in the book. The textual annotations are connected to nodes using red lines and stay connected as the graph layout changes. The toolbar on the upper left is part of the VisDock toolkit and provides tools for annotation, navigation, and selection





# Data Visualization

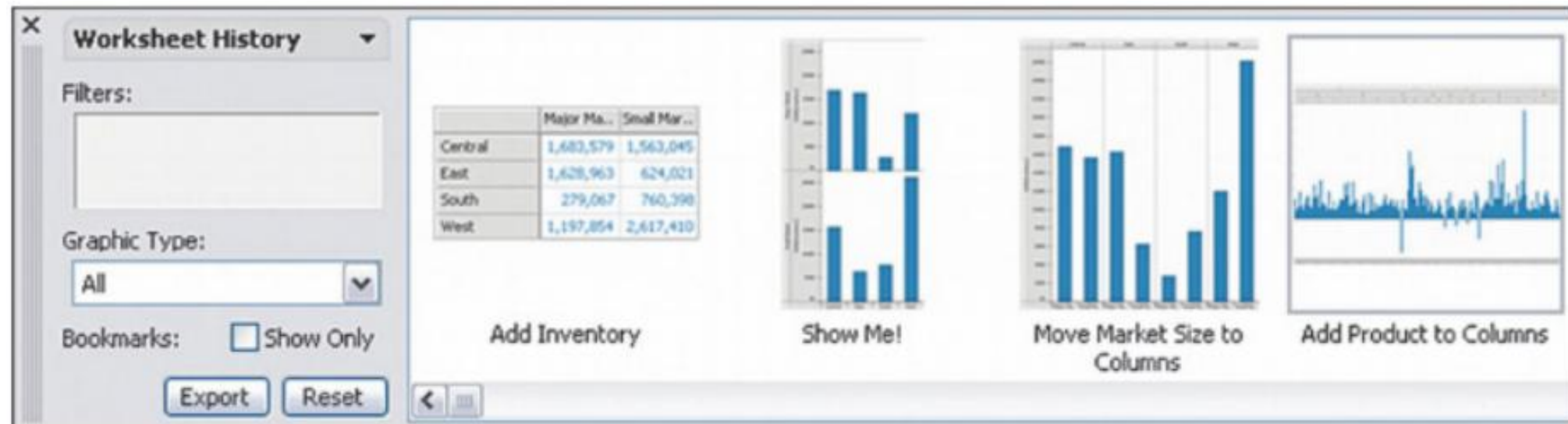
Exploring 284 data breaches in the United States using Keshif (<http://keshif.me/>), a multi-view visualization tool that shows different aspects of the data in separate views (Yalcin, 2016). Selecting items in one view highlights them in others; for example, the user is currently hovering over the bar for “70k–300k” in the view titled “# of Records,” which causes those 124 breaches to also be highlighted in orange in other views, including in the timeline at the bottom.





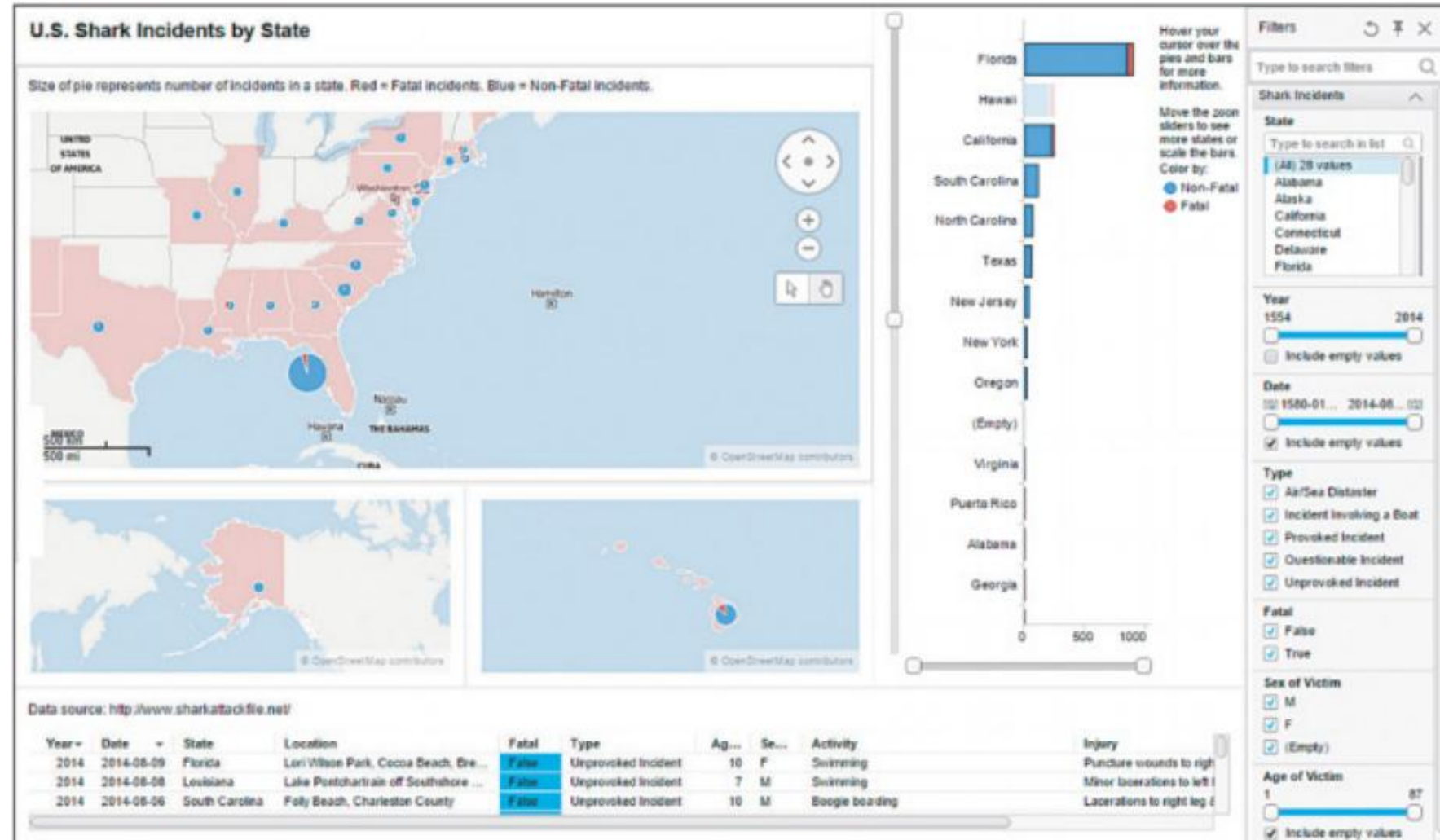
# Data Visualization

Graphical history interface using thumbnails of previous visualization states organized in a comic-strip layout (Heer et al., 2008). The labels describe the actions performed.



# Data Visualization

Spotfire visualization dashboard of shark attacks published on the web. Users can interact with the dashboard, causing views to update dynamically. The tool also allows for application bookmarking (storing the state for specific insights) as well as sharing the analysis on social media platforms such as Facebook, Twitter, and LinkedIn.



# Data Visualization

Web-based visualization of London's 1854 cholera outbreak showing physician John Snow's use of visualization to find its source. This visualization was created in Tableau using its Story Points feature, which allows users to build a narrative from data. The horizontal list of five boxes at the top of the display are the main points in the story, and viewers can be automatically guided through the story by moving to each point from left to right.

## London's 1854 Cholera Outbreak: Data Mapping Halts an Epidemic



He noted that the bulk of deaths were concentrated in an area that generally used the same municipal water pump: Broad Street.

Snow realized that the Broad Street water pump--and poor water quality--was likely the source of the outbreak.

Armed with this information, he went to city authorities.





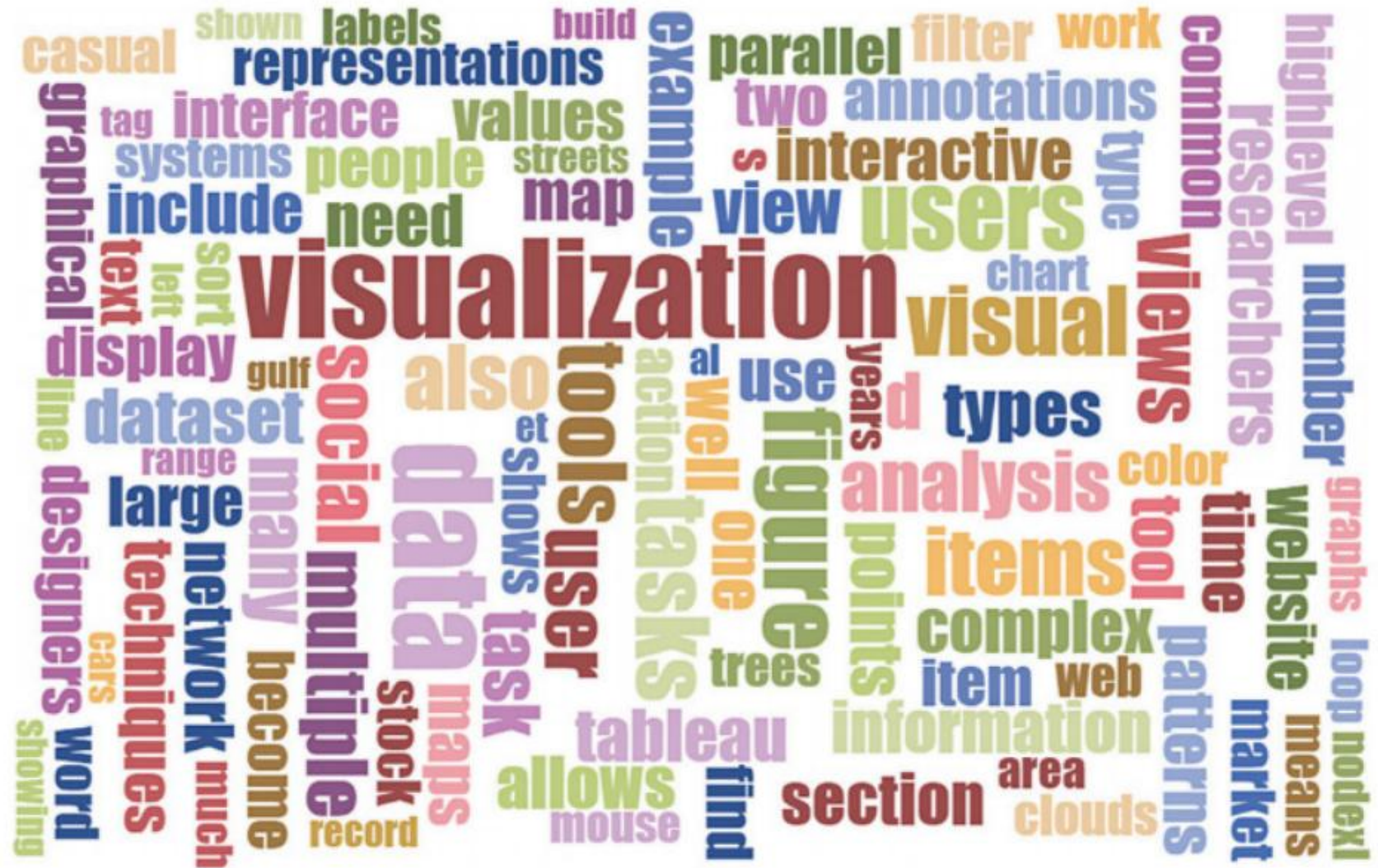
# Data Visualization

- Visualization by Data Type

| <i><b>Data Type</b></i>  | <i><b>Visualization Techniques and Systems</b></i>              |
|--------------------------|-----------------------------------------------------------------|
| <i>1-D linear</i>        | Tag clouds, Wordle, PhraseNets, parallel tag clouds             |
| <i>2-D space</i>         | Geographic information systems (GIS), self-organizing maps      |
| <i>3-D volume</i>        | Volume rendering, medical visualization, molecule visualization |
| <i>Multi-dimensional</i> | Tableau, parallel coordinates, scatterplot matrices             |
| <i>Temporal</i>          | Google Finance, EventFlow, LifeLines, TimeSearcher              |
| <i>Tree</i>              | Treemaps, degree of interest trees, space trees                 |
| <i>Network</i>           | Node-link diagrams, adjacency matrices, NodeXL, Cytoscape       |

# Data Visualization

While tag clouds summarize popular tags used in collaborative tagging applications, word clouds display statistics about word usage in a text collection. Here, a word cloud generated by the online generator at <https://www.jasondavies.com/wordcloud/> shows the most frequent words in this chapter.



# Data Visualization

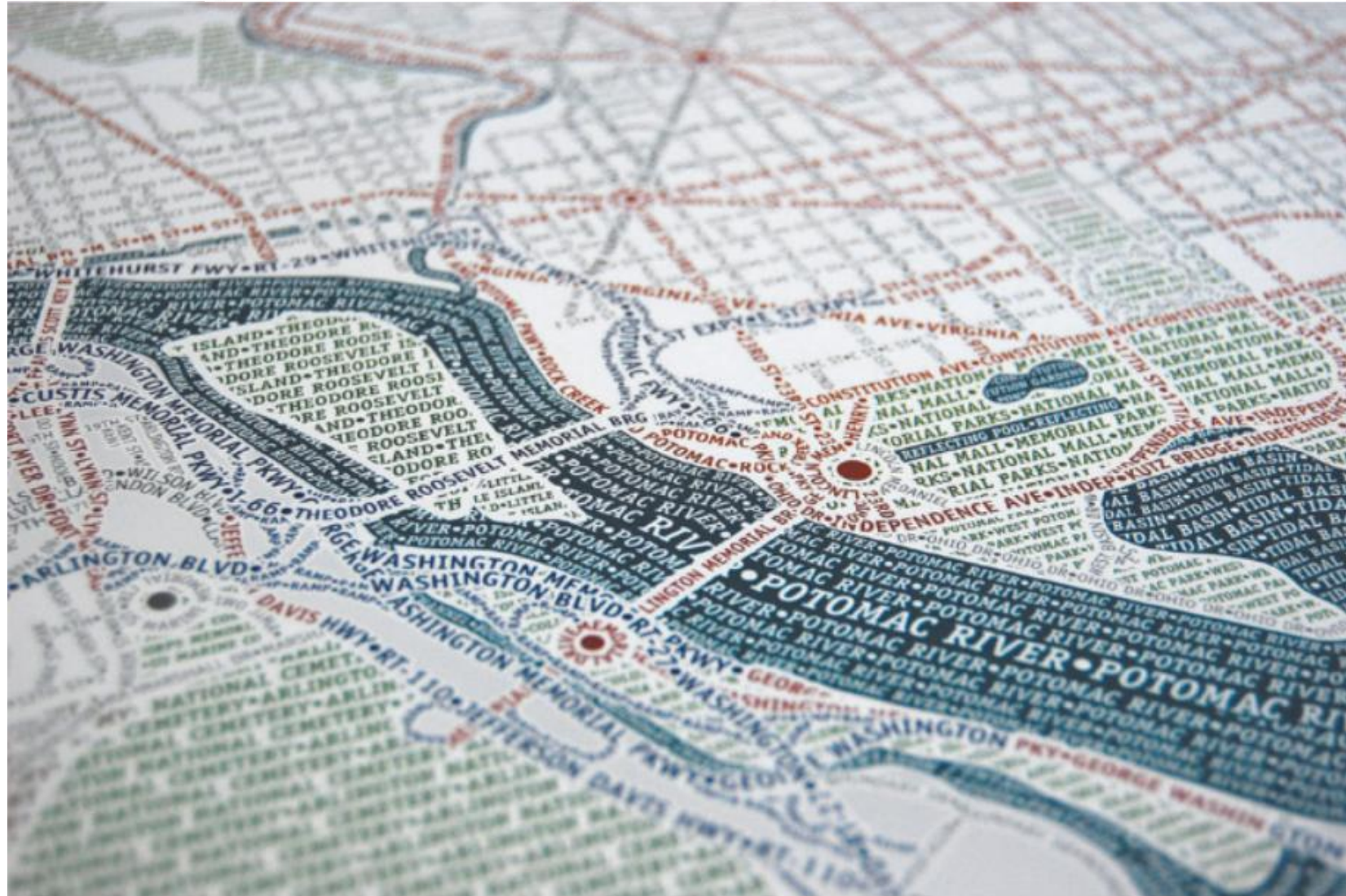
Geographic visualization of gun deaths (suicides versus homicides) in the United States from 2000 to 2014 using data from the Center for Disease Control and Prevention (CDC). Instead of using the actual geographic boundaries of the individual states, this map replaces states with uniform hexagons that have been color-coded using the color scale on the bottom right. The benefit of this representation is to prevent large states from dominating the visual appearance of the overall map. The hexagons have been placed so that they largely preserve the topology of the original map. (<https://public.tableau.com/profile/matt.chambers#!/vizhome/TheGunProblemWeDontMention/TheGunProblemWeDontMention>)





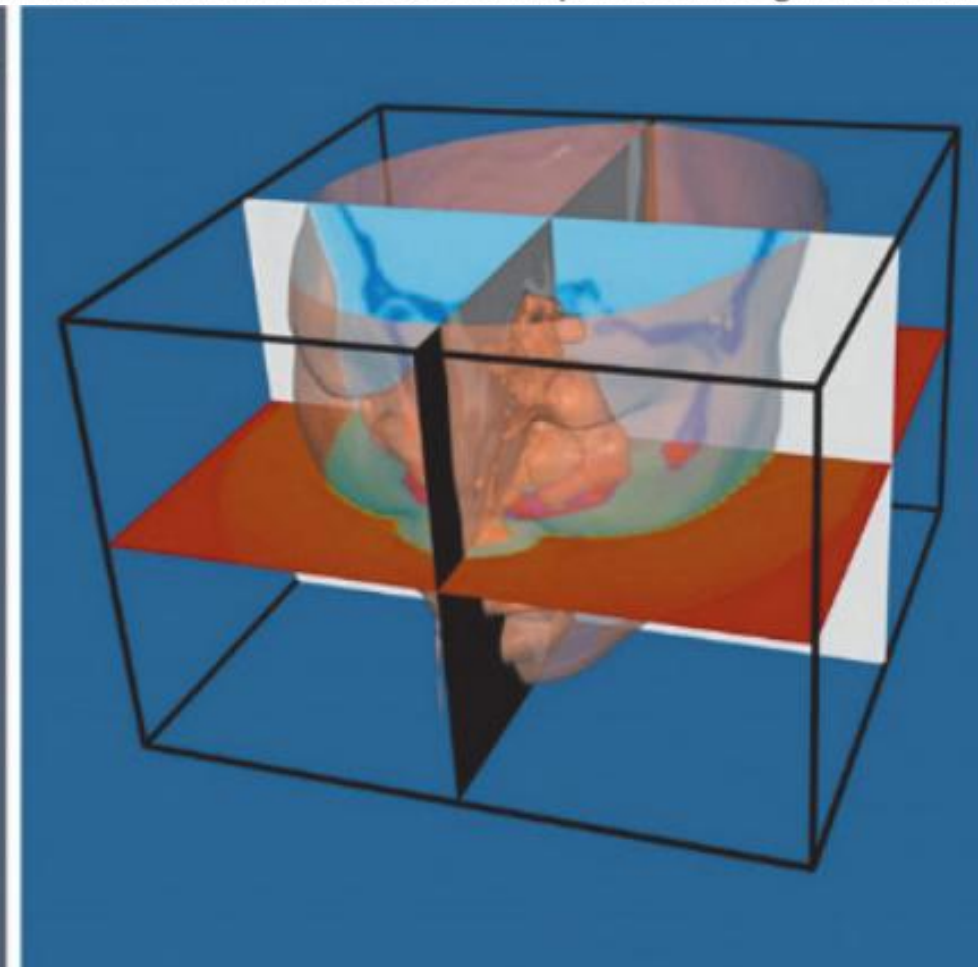
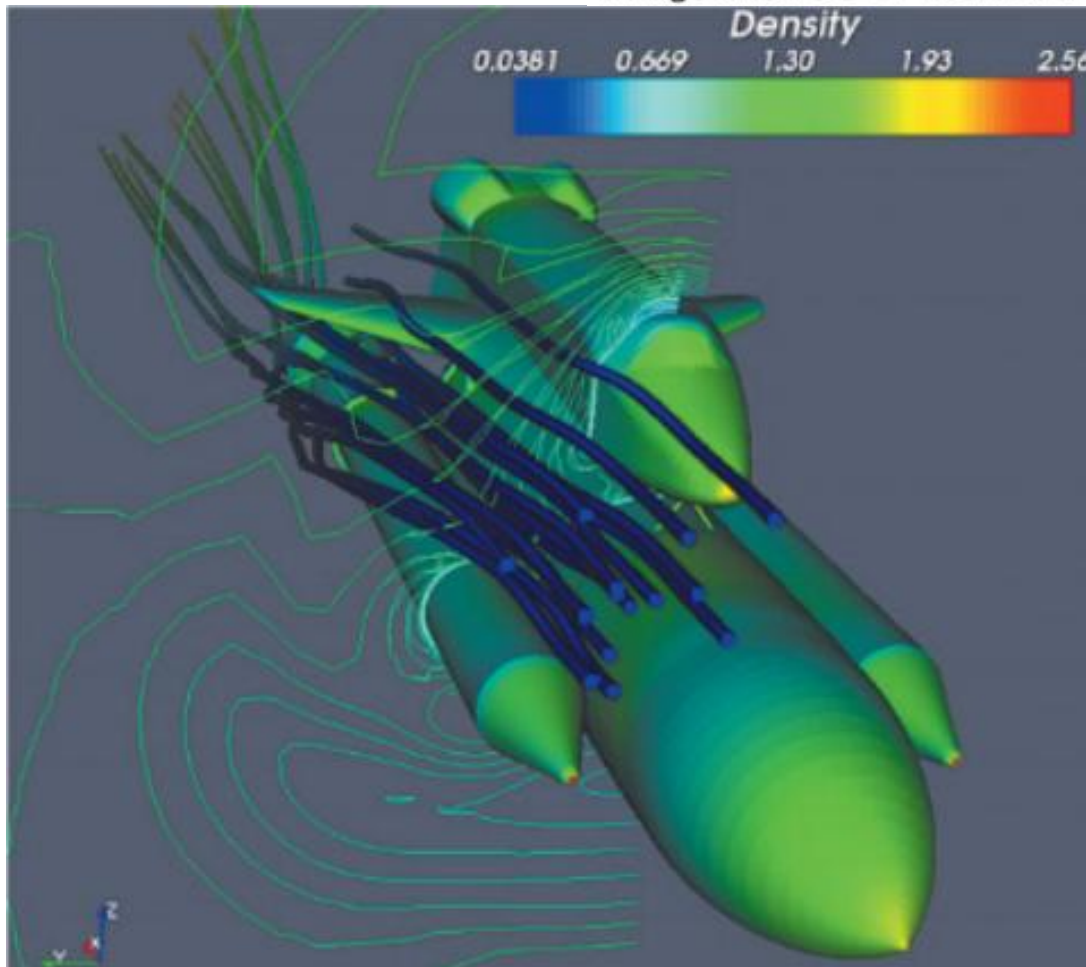
# Data Visualization

Typographic map of Washington, DC, created by Axis Maps. A typographic map consists entirely of text organized into shapes using colored labels of streets, parks, highways, shorelines, and neighborhoods. While this map took a skilled cartographer hundreds of painstaking hours to create, Afzal et al. (2012) later proposed an automatic approach taking mere minutes.



# Data Visualization

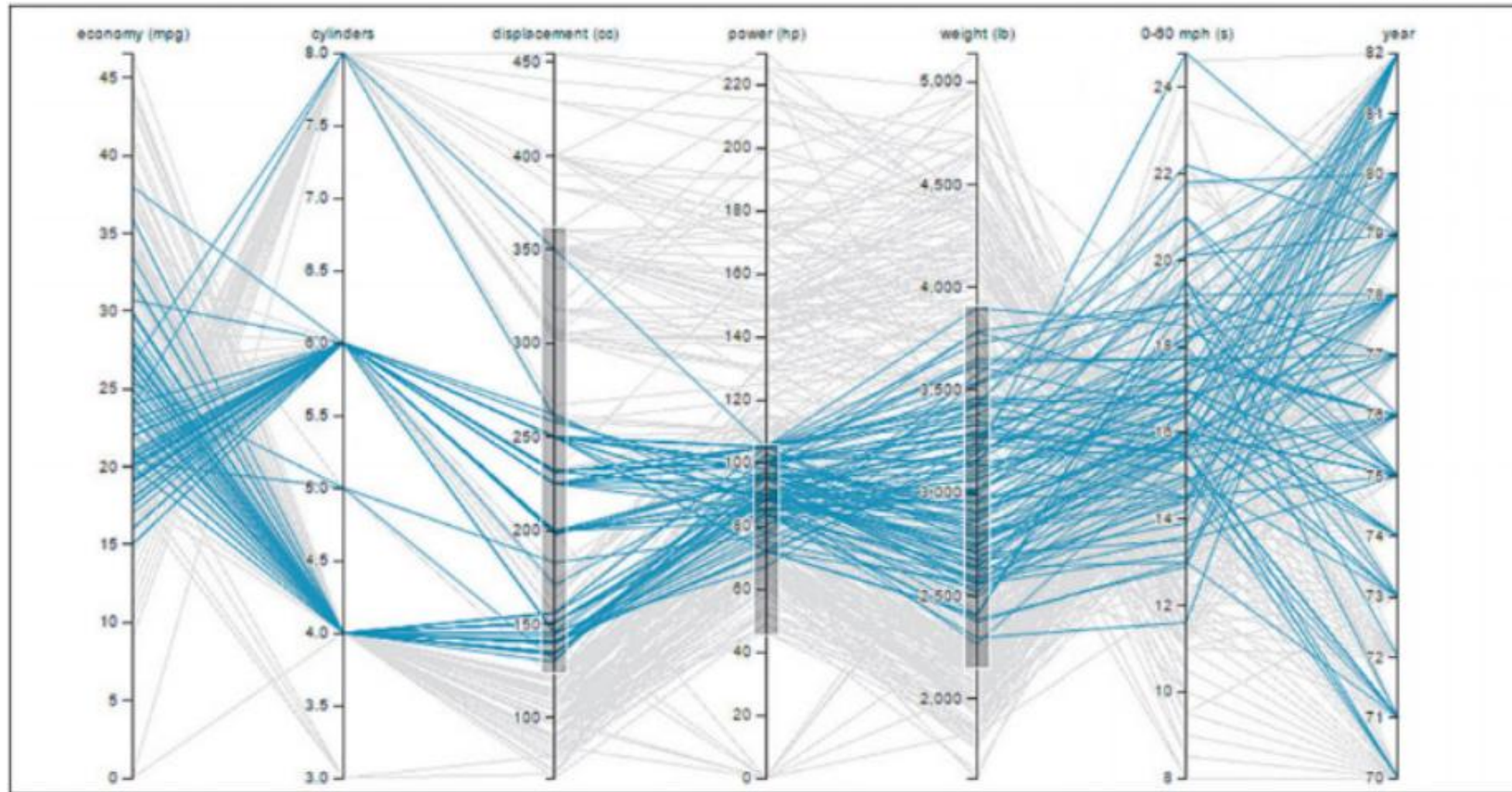
Two 3-D visualizations created using the Visualization Toolkit (VTK), a commercial software development library by Kitware, Inc. (<http://www.kitware.com/>). The left image shows flow density around the space shuttle using a rainbow color scale. The right image shows a CT scan of a human head with cross-sectional planes through the data.





# Data Visualization

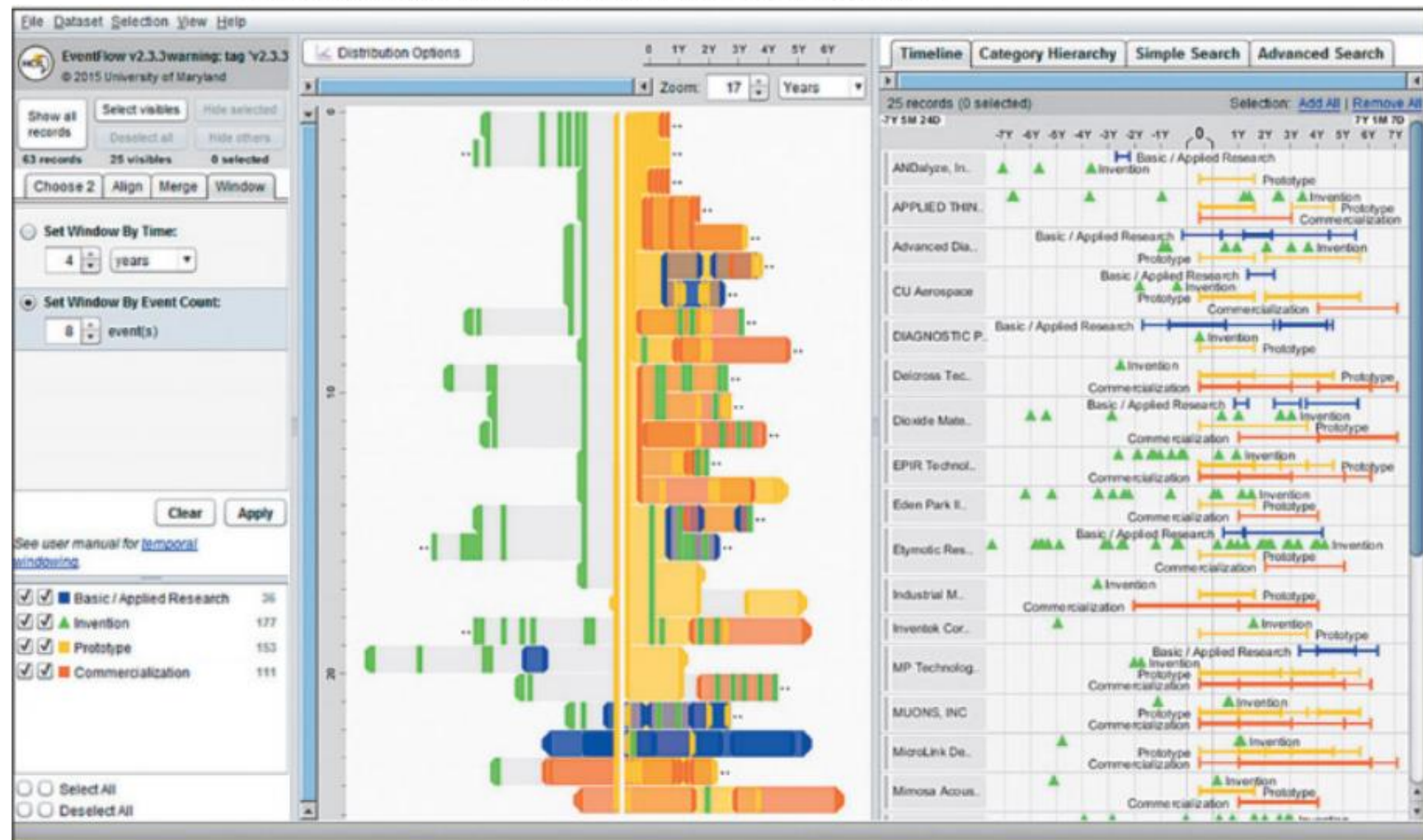
Parallel coordinate visualization of cars from the 1970s and 1980s created using the D3 library. This visualization supports axis filtering where selecting data ranges on the dimension axes filters out the cars that do not meet all of the criteria (gray lines).





# Data Visualization

The EventFlow (<http://www.cs.umd.edu/hcil/eventflow>) temporal event visualization system used to visualize sequences of innovation activities by Illinois companies. Activity types include research, invention, prototyping and commercialization. The timeline (right panel) shows the sequence of activities for each company. The overview panel (center) summarizes all the records aligned by the first prototyping activity of the company. In most of the sequences shown here, the company's first prototype is preceded by two or more patents with a lag of about one year between the last patent application and the first prototype.



# Data Visualization

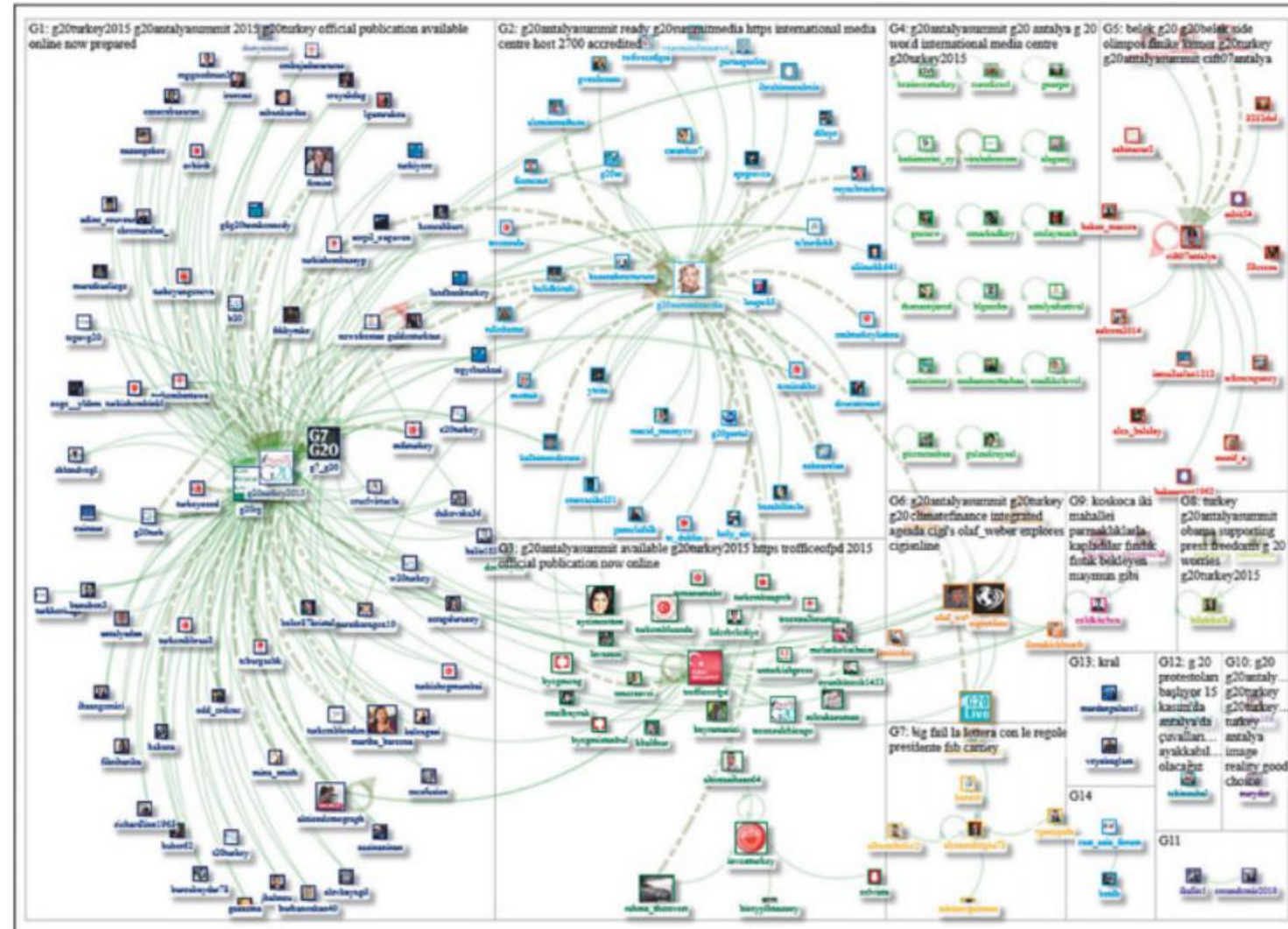
The S&P 500 Market Monitor by Visual Action (<http://www.visualaction.com/>) a web-based treemap visualization showing stock performance of the 500 large companies making up the S&P 500 index on the NYSE and NASDAQ stock markets. Each rectangle represents a company, sized according to its market capitalization, colored based on its one-day change, and organized into sectors.





# Data Visualization

Social network visualization built using NodeXL (Hansen et al., 2010) of 191 Twitter users tweeting with the hashtag “#G20AntalyaSummit” on November 9, 2015. The hashtag refers to the 2015 G-20 summit held in Antalya, Turkey, on November 15–16, 2015. The users have been grouped and laid out in boxes based on the contents of the tweets. NodeXL (<https://nodexl.codeplex.com/>) allows social scientists to collect, analyze, and visualize network graphs using a familiar interface that plugs into Microsoft Excel.





DISCUSSION

Q & A